

NEX

Network of Engineering Xcellence

Organizational Profile

Prepared for prospective partners, sponsors, and collaborators

1. Introduction

Network of Engineering Xcellence (NEX) is an interdisciplinary innovation society established to unite students across the Faculty of Engineering, the Faculty of Agriculture, and the Faculty of Environmental Sciences around a common purpose: to identify, research, develop, and implement practical solutions to real-world problems through collaboration, innovation, and excellence.

NEX exists because the world's greatest challenges cannot be solved by one discipline alone. Problems involving food security, renewable energy, waste management, healthcare, climate change, infrastructure, manufacturing, artificial intelligence, and sustainability require engineers and scientists from different backgrounds working together. NEX provides that platform.

At the center of everything NEX does is one commitment: we do not stop at ideas. Our primary focus is developing real, working solutions to real problems — taking each challenge from research through to a built, tested, and documented outcome.

2. Our Founder

NEX was conceived and founded by Oyewole Samod Atanda, a first-class graduate of Chemical and Polymer Engineering from Lagos State University, a multiple-time recipient of the Scholar of the Year award. Driven by a conviction that engineering education must extend beyond the classroom into real, implemented solutions, he conceptualized NEX and entrusted its development and execution to the President, working closely together to bring the vision to life. NEX stands today as the realization of that founding idea — a testament to what happens when academic excellence is paired with a genuine commitment to impact.

3. The Problem We Address

Universities produce brilliant students, but many ideas never leave the classroom. Research often ends as a report. Competitions end after the awards are given out. Projects rarely become implementation. Industries consistently report that graduates lack practical, hands-on experience, while communities continue to face the same unresolved problems year after year.

NEX was created to bridge this gap — to ensure that the knowledge and talent inside the university is actually converted into solutions that work in the real world.

4. Our Vision

To become the leading student-driven interdisciplinary innovation ecosystem that transforms ideas into impactful solutions for society.

5. Our Mission

To bring together students across Engineering, Agriculture, and Environmental Sciences to:

- Collaborate across disciplines
- Conduct rigorous, evidence-based research
- Develop innovative technologies

- Build working prototypes
- Implement practical solutions
- Document knowledge thoroughly
- Create lasting societal impact

6. Our Philosophy

NEX believes engineering education should go beyond lectures. Students should learn, research, collaborate, build, test, implement, document, and improve — a complete cycle that defines genuine innovation. Every NEX project is expected to follow this full pipeline, from problem identification through to publication and continuous improvement, rather than stopping at a presentation or a prototype that never leaves the lab.

7. What Makes NEX Different

NEX is not a departmental society, a political organization, or a group that exists only to hold meetings. NEX is an innovation ecosystem, and every project follows a complete pipeline:

- Problem Identification
- Research
- Brainstorming
- Prototype Development
- Testing
- Implementation
- Impact Assessment
- Documentation
- Publication
- Continuous Improvement

Most student organizations stop after the presentation stage. NEX exists to finish the cycle — and building a working solution to a real problem is the central, non-negotiable output of every NEX initiative.

8. Focus Areas

NEX directs its projects, or "Missions," toward real societal challenges across six focus areas:

- Water — purification, irrigation, flood management, distribution systems
- Energy — solar, bioenergy, biogas, energy efficiency
- Waste — recycling, circular economy, waste-to-energy, plastic management
- Agriculture — smart farming, food security, post-harvest preservation, precision agriculture
- Infrastructure — affordable housing, smart cities, sustainable construction
- Digital Innovation — artificial intelligence, software development, automation, IoT, robotics

9. Core Activities

NEX operates through a consistent set of activities designed to keep the innovation pipeline moving year-round:

- Semester-Based NEX Missions — each semester, members tackle one major real-world challenge (e.g. food security, campus waste management, renewable energy) as a structured, multi-week project
- Workshops — covering AI, research methods, programming, CAD, embedded systems, design thinking, scientific writing, and entrepreneurship
- Hackathons — interdisciplinary competitions, including industry-set challenges and prototype development sprints

- Research Groups — collaborative work on research papers, technical reports, and innovation projects
- Community Outreach — working directly with communities to understand problems, develop solutions, implement projects, and measure real impact

10. The Role of Research

Research is the heartbeat of NEX. Every solution developed under the NEX name is expected to be evidence-based, well-documented, repeatable, and measurable. NEX exists in part to revive a culture where research leads to implementation, rather than ending in an archive.

11. Long-Term Goals

Within five years, NEX aims to:

- Become the leading interdisciplinary innovation society on campus
- Develop implementable engineering solutions to real community problems
- Publish annual research and innovation reports
- Host flagship innovation challenges
- Build sustained partnerships with industry and academia
- Secure grants to fund student-led projects
- Expand to other universities

12. Core Values

- Excellence
- Innovation
- Integrity
- Collaboration
- Leadership
- Professionalism
- Impact
- Sustainability
- Service

13. Leadership Philosophy

Leadership in NEX is understood as service. Executives are expected to mentor members, build lasting systems, encourage collaboration, promote excellence, and leave the organization stronger than they found it.

14. Organizational Culture

NEX promotes a culture where ideas are welcomed, questions are encouraged, and failure is treated as part of learning. At the same time, execution matters more than promises, research is celebrated, innovation is expected, documentation is mandatory, and impact is measurable.

15. Executive Structure

NEX is led by a nine-member executive team, each responsible for a distinct function of the organization. Roles are structured to ensure the full innovation pipeline — from research to implementation to communication and partnerships — is actively managed.

President

Overall head of NEX; sets strategic direction and holds final accountability for the organization.

- Provides overall vision and strategic leadership for the society
- Chairs executive meetings and coordinates the executive team
- Represents NEX in official and public capacities, including partnership discussions
- Ensures the organization's mission, values, and innovation pipeline are upheld across all activities
- Holds final decision-making authority on major organizational matters

Vice President

Deputizes for the President and drives execution of key initiatives.

- Supports the President in strategic planning and day-to-day leadership
- Steps in for the President when required
- Drives cross-functional initiatives and ensures alignment across departments
- Champions the development and execution of practical, high-impact solutions

Technical & Research Lead

Oversees the technical integrity and research rigor of every NEX project.

- Guides members through the research phase of the innovation pipeline
- Reviews and approves research submissions and technical approaches
- Ensures projects are evidence-based, well-documented, and technically sound
- Provides technical mentorship across focus areas and disciplines

General Secretary & Operations

Manages the internal administration and day-to-day operations of NEX.

- Maintains official records, minutes, and organizational documentation
- Coordinates the smooth running of meetings, sessions, and events
- Tracks operational execution across departments and projects
- Ensures administrative processes run efficiently and reliably

Programs & Events Lead

Plans and delivers NEX's workshops, missions, and events.

- Designs and schedules the workshop and events calendar
- Coordinates semester-based NEX Missions from planning to execution
- Organizes hackathons and other structured programs
- Ensures programs run smoothly and deliver on their intended learning or project outcomes

Head of Digital Infrastructure & Web Systems

Media and Communications Lead; owns NEX's digital platforms and technical infrastructure.

- Leads development and maintenance of the official NEX website and digital tools
- Ensures the digital ecosystem supports registration, project tracking, and member engagement
- Maintains the technical reliability and security of NEX's digital infrastructure
- Works closely with the Communications & Media team to ensure digital platforms reflect NEX's brand and content strategy

Communications & Media Officer

Executes NEX's public-facing content and communications strategy.

- Produces and publishes content across NEX's communication channels
- Designs visual assets, including flyers and promotional material
- Maintains consistent messaging and brand identity across platforms
- Supports publicity for events, missions, and organizational milestones

Assistant Communications & Media Lead

Supports the communications function and adds versatile content and design capacity.

- Assists in content creation, graphic design, and documentation of NEX activities
- Supports the production of presentations, reports, and promotional materials
- Helps maintain a consistent and professional public image for NEX
- Provides backup coverage across communications tasks as needed

External Relations & Partnership Lead

Builds and manages NEX's relationships with outside organizations.

- Leads outreach to universities, research institutes, companies, NGOs, government agencies, and innovation hubs
- Manages ongoing partnership relationships and communication
- Represents NEX in external meetings, negotiations, and events
- Identifies new opportunities for funding, mentorship, internships, and collaborative projects

Membership & Community Lead

Owens the member experience and community growth of NEX.

- Oversees member onboarding, engagement, and retention
- Coordinates community outreach initiatives connected to NEX Missions
- Serves as a point of contact for member concerns and feedback
- Supports the growth and health of NEX's overall community

Finance Officer

Manages the financial resources of NEX.

- Oversees budgeting, expense tracking, and financial planning for NEX activities
- Manages funds received through partnerships, sponsorships, or grants
- Ensures transparent and responsible use of organizational resources
- Provides financial reporting to the executive team as needed

16. Our Guiding Principle

Research should not end in reports. Engineering should not end in classrooms. Innovation should not end in competitions. Every idea should strive to create measurable impact through collaboration, implementation, and continuous improvement.

17. Our Identity

NEX is more than a society. It is a network, a research community, an innovation hub, a problem-solving ecosystem, and a leadership academy — a platform where students transform knowledge into impact.

NEX exists to awaken a culture of research, innovation, and implementation within the university by empowering students to collaborate across disciplines, build practical solutions, document their work, and create lasting impact in society.